Project 3: Exploring Keyword Co-occurrence Networks in Scientific Publications

1. Aims and Objective

The main objective of this project is to analyze the dataset of scientific publication and transform keyword details into a structured co-occurrence network. Here, different scientific publications datasets were studied forming a structured network from the lists of keywords. With the calculated network metrics like Degree and Strength, the project seeks to find influential keywords which can acts as central hubs and to understand how different research area are connected in border academic fields.

2. Methodology

- The keywords columns designated as Keywords 1 to Keywords 12 were from the provided dataset were normalized to lowercase and whitespace were removed.
- The unique keywords were then paired from each article.
- The symmetric weighted adjacency matric was created.
- Using the Network library, the undirected graph was then created to obtain Degree and Strength.
- Finally, for visualizations, a scatter plot is created to support analysis and interpretation.

3. Results

The obtained network contains 248 nodes and 2,141 edges. In other words, the network contains 248 keywords and 2141 unique connections.

Top Ten Nodes by Degree

Rank	Keyword	Degree
1	organizational behavior	166
2	organizational effectiveness	104
3	management science	102
4	personnel management	93
5	decision making	90

6	organizational structure	74
7	organizational sociology	66
8	strategic planning	66
9	industrial management	64
10	corporate governance	62

Top Ten Nodes by Strength

	Keyword	Strength
1	organizational behavior	265
2	organizational effectiveness	144
3	management science	136
4	personnel management	126
5	decision making	112
6	organizational structure	107
7	organizational sociology	96
8	strategic planning	80
9	industrial management	84
10	corporate governance	85

Top nodes pairs by Weight

Rank	Keyword 1	Keyword 2	Weight
1	Organizational Behavior	Organizational Effectiveness	11
2	Organizational Behavior	Organizational Structure	9
3	Organizational Behavior	Personnel Management	8
4	Management Science	Organizational Behavior	7

5	Organizational Effectiveness	Organizational Structure	6
6	Organizational Behavior	Organizational Sociology	6
7	Decision Making	Organizational Behavior	6
8	Corporate Governance	Organizational Behavior	6
9	Organizational Sociology	Teams in the Workplace	5
10	Industrial Management	Organizational Behavior	5

Degree and Strength relationship

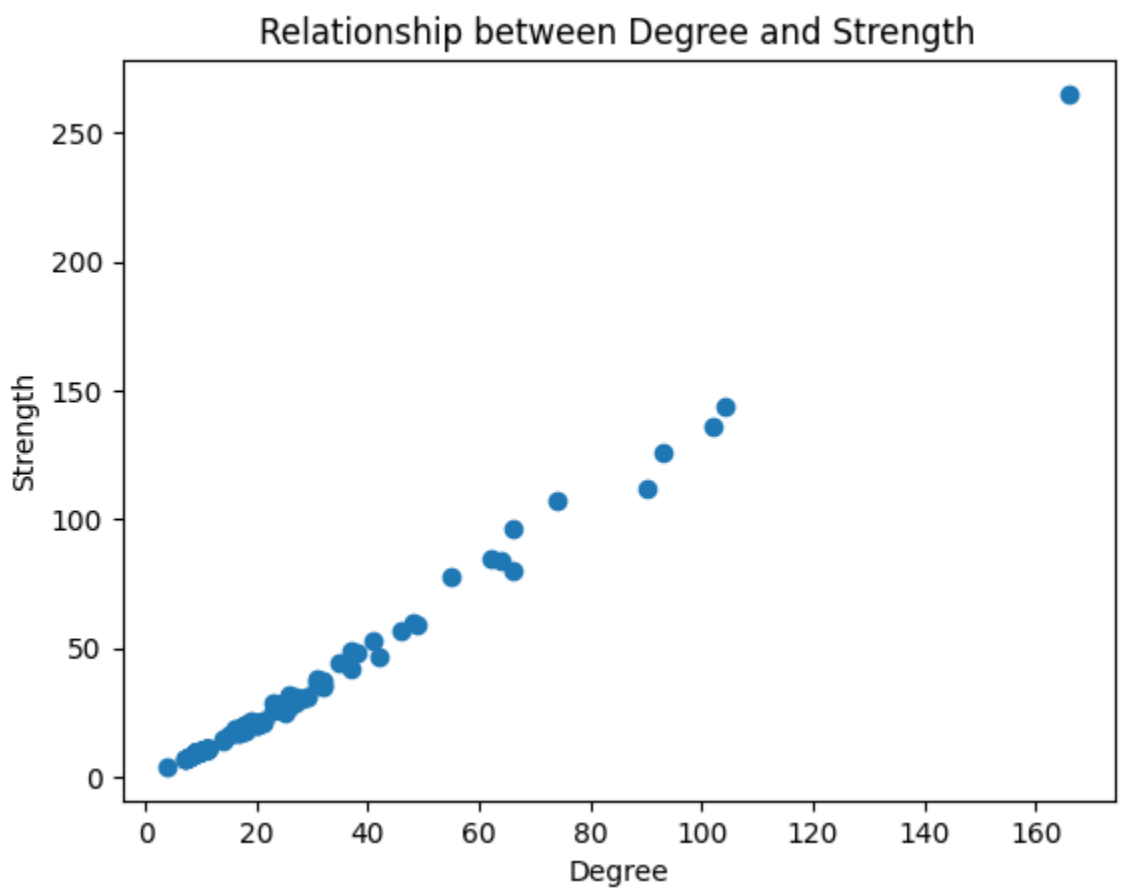

Figure 1: Scatter plot showing relationship between Keyword Degree and Strength.

As per the results obtained, we can conclude that the keyword Organizational Behavior is the most important node in the entire network. The keyword has degree of 166 and strength of 265 meaning that it is the most discussed topic. Also, the most frequent co-occurrence pair is “organizational behavior and “Organizational effectiveness”. This pair has a weight of 11.
As we also can see from the scatter plot (Figure: 1) that the strong positive relationship between Strength and degree is linear with correlation value of approx. 0.99. With this we can conclude that keywords that connect to many other keywords also appear to be more frequent (high strength).
In conclusion, we can say that the keyword network is well-structured and organized.

4. Conclusion

The network analysis reveals the structural framework of the provided publication data. The strong connections around organizational behavior indicate that it is a key linking topic that connects various areas of management and data analytics. Overall, this analysis shows how network methods can reveal core research structures, identify influential topics and suggest better academic and strategic decision- makings.